bounce. Does a true singularity occur? Or does the bounce happen without the ultimate maximum density being be reached, allowing for the possibility that information of some kind could survive the event and pass on into the next cycle? The latest version of the model has very little contraction, so nothing like a singularity occurs. Instead of using a collision between branes, the contraction in that model is driven by a *scalar field*, which is something similar to the Higgs field, or (possibly) to what we think might have caused inflation. That model does offer the tantalizing possibility that information might pass between cycles, and we could in principle someday see evidence of that.

Which brings us to the question of observational evidence. Since both the ekpyrotic model and inflation were designed to solve the same cosmological problems, it may take a little creativity to confirm or rule out either of them. Everything we've seen so far in the cosmos seems compatible with the standard inflation picture, but we haven't seen a smoking gun for it, nor have we seen anything that proves or kills off an ekpyrotic alternative. Arguments have gone back and forth for years about whether cyclic models are more or less *theoretically* appealing than inflation, but observationally it's still an open question. Some data to finally decide the issue would be extremely helpful here.

Our best bet might be finding evidence for *primordial gravitational waves*: large-scale ripples in space that would originate not from merging black holes or neutron stars but from the violence of the inflationary epoch, when the first seeds of cosmic structure were laid down by quantum wiggles in the inflaton field. If found, these would be as close as we're likely to get to the elusive smoking gun for inflation. Briefly, in 2014, the cosmology community lit up with excitement as the leaders of an experiment called BICEP2[fn7] announced that they'd seen evidence for exactly that. Looking at the polarization of the light from the cosmic microwave background, they saw what appeared to be twisting patterns that could only have come from gravitational waves distorting space during the epoch of primordial fire. These patterns were held up as a discovery so revolutionary Nobel Prizes were virtually guaranteed. After all, even setting aside the implications for inflation, they were a solid observation of gravitational waves (more than a year before LIGO saw its first black holes collide) *and*, because of the connection to quantum wiggles, they were the first evidence of any kind for the quantum nature of gravity.